KHULNA CITY CORPORATION

Smart Sweep

The city's waste collection in one system — from the first knock on a door to the month's bill.

Progress update · October 2026

Where we stand

Every part of the system is built and working. Twelve screens, one database, nothing left as a mock-up.

Built

All twelve modules are finished and feed each other.

In Bangla

Bangla by default, English on a switch, Bangla numerals throughout.

Works offline

A collector with no signal keeps working; the phone uploads later.

Figures in this deck come from the loaded pilot dataset, not yet from city-wide operations.

What it replaces

The work was always being done. What was missing was a record of it that anyone could check.

ON PAPER, BEFORE	IN SMART SWEEP, NOW
A register of households in a ledger	A searchable list, each home QR-tagged and pinned on a map
No record of which homes were served today	Every visit timed and attributed to a collector
Bills written and chased by hand	Bills raised monthly, payments matched, arrears listed
Complaints taken by phone, then forgotten	A ticket with a clock, a photo and an owner
Figures assembled by hand for each meeting	The same figures on screen, always current

One cycle, six steps

1

Survey

A home is recorded at the door

2

Register

It joins the customer list

3

Route

It is added to a round

4

Collect

The van calls; the visit is logged

5

Bill

A monthly bill, then the payment

6

Report

The month adds up on its own

Each step hands the next one what it needs. Nobody re-types anything, and no step can be skipped — a home with no confirmed location cannot be put on a round.

Twelve screens, all working

Dashboard

The month at a glance, by zone

Live map

Vans moving on a map of Khulna

Households

The customer register, QR-tagged

Collection

A collector's round for the day

Routes

Stop by stop, as the walk happens

Route planning

Build a round and assign it

Surveys

The door-to-door questionnaire

Complaints

Tickets with a clock and a photo

Billing

Fleet

Collectors

Reports

Finding the homes that pay nothing

A surveyor walks a ward with a 73-question form and records every premises — including the ones the city has never billed.

The survey is kept

When a surveyed home becomes a paying customer, its survey stays attached. You can see how many were found, how many were approved, and how many now pay.

A supervisor approves

Nothing a surveyor writes reaches the billing register on its own. A supervisor checks it, and only an administrator can turn it into a charge.

Set up for Khulna this month: survey area, district, thana, date and partner organisation now fill themselves in, so a surveyor answers five fewer questions at every door.

A collector's day

Signs in with a PIN

Four digits. No password to remember, no typing in the rain.

Scans the QR tag

The tag on the gate says which home this is. No list to search.

Marks it collected

Or skipped, with a reason: gate locked, nobody home, refused.

Needs no signal

The round is saved on the phone and sent up when the signal returns.

If part of an upload fails, the system names exactly which homes to retry — a morning's work is never silently lost.

The money, counted by the system

372

Bills raised

One run covers every household

280

Payments taken

Full or part payment, both recorded

69%

Collected

Of everything billed in the pilot

51

Overdue

Named, listed, ready to chase

Who has not paid, and how much, is now one screen instead of a week’s work.

Pilot dataset, October 2026.

Complaints with a clock on them

Every complaint has an owner, a deadline and a photograph — and a record of who did what to it.

Raised

By phone, in person or from the field. Logged the same way whichever.

Timed

The clock starts at once, and a ticket near its deadline is flagged.

Evidenced

Photographs attach to the ticket, before and after.

Closed

With a reason, visible to whoever asks about it months later.

What the Corporation sees

The dashboard

Households served, waste by zone, the last seven days, open complaints — on one screen, current.

The live map

Where the vans are right now, moving on a map of Khulna, with the round each one is working.

The reports

Service, revenue, arrears and reconciliation — exported to Excel, CSV or PDF for a meeting.

Each person sees only their own area. A ward collector sees their round, a supervisor their zone, the Corporation the whole city.

Built for the conditions here

Bangla first

The whole system is in Bangla by default, Bangla numerals included. English is a switch for anyone who prefers it.

A phone, not a computer

Collectors work on the phone already in their pocket. Sign in with a PIN, scan a tag, done.

Unreliable signal assumed

Nothing depends on being online at the moment it happens. Work is saved and sent when it can be.

Two contractors, kept apart

Each agency sees only its own staff, households and bills. The system enforces that, not a policy document.

What comes next

1

A survey app for the phone

Surveyors still work from the web screen. A phone app would let them work at the door with no signal at all, and photograph the premises. The groundwork is done; the app itself is not built.

2

Two gaps to close first

A surveyor's own account cannot yet submit a survey, and photographs can only be added by an administrator. Both are small changes, and both block the phone app.

3

Widen the pilot

The system is loaded with four wards. Adding the rest is data entry and survey work, not development.